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1. Adjacency Matrix

① create graph GraphMatrix graph = new GraphMatrix(4);
creates a 4x4 matrix with 0.

② add edges

graph.addEdge(0,1) → connects 0 & 1
graph.addEdge(0,2) → connects 0 & 2
graph.addEdge(1,3) → connects 1 & 3
graph.addEdge(2,3) → connects 2 & 3

graph is undirected so [source][destination] & [destination][source] become 1

vertex	0	1	2	3
0	0	1	1	0
1	1	0	0	1
2	1	0	0	1
3	0	1	1	0

Adjacency matrix:

0 1 1 0
1 0 0 1
1 0 0 1
0 1 1 0

2. BFS using adjacency list (without collections)

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Edges : 0-1, 0-2, 1-3, 1-4, 2-5
- Start vertex 0
- queue initially : [0]
- visit 0 → add 2,1
visit 2 → add 5
visit 1 → add 4,3
visit 5 → no new vertex
visit 4 → no new vertex
visit 3 → no new vertex

o/p BFS traversal (Adjacency list) : 0 2 1 5 4 3

3. BFS using adjacency list (collections)

Start vertex = 5
visit 5 → add 2
visit 2 → add 0
visit 0 → add 1
visit 1 → add 3, 4
visit 3 → no new vertex
visit 4 → no new vertex

o/p output - BFS traversal : 5 2 0 1 3 4

4. BFS using Adjacency matrix (without collections)

Start vertex = 0

Queue initially : [0]

Visit 0 → add 1, 2

Visit 1 → add 3, 4

Visit 2 → add 5

Visit 3 → no new vertex

Visit 4 → no new vertex

Visit 5 → no new vertex

output - BFS traversal : 0 1 2 3 4 5